



Experiment 8 (Data Preprocessing)

Aim: Perform Data cleaning on a given dataset.

Theory:

Data cleaning is the process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset. When combining multiple data sources, there are many opportunities for data to be duplicated or mislabeled. If data is incorrect, outcomes and algorithms are unreliable, even though they may look correct. There is no one absolute way to prescribe the exact steps in the data cleaning process because the processes will vary from dataset to dataset.

After cleansing, a data set should be consistent with other similar data sets in the system. The inconsistencies detected or removed may have been originally caused by user entry errors, by corruption in transmission or storage, or by different data dictionary definitions of similar entities in different stores. Data cleaning differs from data validation in that validation almost invariably means data is rejected from.

Missing values are usually represented in the form of Nan or null or None in the dataset. **df.info()** the function can be used to give information about the dataset. This will provide you with the column names along with the number of non – null values in each column.

The second way of finding whether we have null values in the data is by using the **isnull()** function. Also, calculate the ratio of null values to the total values and drop those columns whose ratio is > 0.5 .

The **describe()** method returns description of the data in the DataFrame. If the DataFrame contains numerical data, the description contains this information for each column.

df.rename() will be used to rename a column to any specific column name.

You can drop a column using **dropna(axis=1)** for columns and replace **axis=0** for rows to drop the column and rows with 'NaN' values.

df.duplicated() returns a series with True and False values that describe which rows in the DataFrame are duplicated and not.

The **fillna()** function to fill the null values in the dataset. The values can be filled with either 0 or mean of the data. Crude Analysis can be used if you need to analyse certain columns based on criteria in which relationship of one variable with other.

The date field is an important attribute that needs to be handled during the cleansing of data. There are multiple different formats in which data can be entered into the dataset. Therefore, standardizing the date column is a critical task. Some people may have treated the date as a string



column, some as a DateTime column. When the dataset gets combined from different sources then this might create a problem for analysis. Visualization of various columns can be done using a bar graph, histogram or any appropriate graphs.

- Remove duplicate or irrelevant observations
- Fix structural errors
- Filter unwanted outliers
- Handle missing data
- Validate

Lab Assignment to complete:

Dataset: Employee-turnover.excel

Implement the following on the dataset given above:

1. Find the missing values and replace it with 0 or mean.
2. Dropping Columns in a DataFrame.
3. Renaming Columns and Skipping Rows.
4. Make age bins of 0-20, 20-40, 40-60 from age column.
5. Date Transformation Operations.
6. Visualization of Attrition column based on the correlation with the help of a heatmap.
7. Out of those columns in q.6 which are highly correlated. (Consider 2-3 for example)



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FDAexp08.ipynb

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Files

sample_data
Employee-turnover (1) (2).xlsx

Code

Text

RAM

Disk

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```
[4] import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_excel("/content/Employee-turnover (1) (2).xlsx")
df.head()
```

	Employee Count	Employee ID	Department	Job Role	Gender	Age	Marital Status	Education	Education Field	Business Travel	Total Working Years	Years At Company	Years In Current Role	Years Since Last Promotion	Years Since Last Managerial
0	1.0	1.0	Sales	Sales Executive	Female	41.0	Single	College	Life Sciences	Travel_Rarely	8.0	6.0	4.0	0.0	
1	1.0	2.0	Research & Development	Research Scientist	Male	49.0	Married	Below College	Life Sciences	Travel_Frequently	10.0	10.0	7.0	1.0	
2	1.0	4.0	Research & Development	Laboratory Technician	Male	37.0	Single	College	Other	Travel_Rarely	7.0	0.0	0.0	0.0	
3	1.0	5.0	Research & Development	Research Scientist	Female	33.0	Married	Master	Life Sciences	Travel_Frequently	8.0	8.0	7.0	3.0	
4	1.0	7.0	Research & Development	Laboratory Technician	Male	27.0	Married	Below College	Medical	Travel_Rarely	6.0	2.0	2.0	2.0	

5 rows x 29 columns

```
[6] df.info()
```

React App

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```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 29 columns):
 #   Column                Non-Null Count  Dtype
---  --
 0   Employee Count         1470 non-null   float64
 1   Employee ID            1470 non-null   float64
 2   Department              1470 non-null   object
 3   Job Role                1470 non-null   object
 4   Gender                  1470 non-null   object
 5   Age                     1470 non-null   float64
 6   Marital Status          1470 non-null   object
 7   Education                1470 non-null   object
 8   Education Field          1470 non-null   object
 9   Business Travel          1470 non-null   object
10   Distance From Home (kms) 1439 non-null   object
11   Job Involvement          1470 non-null   object
12   Job Level                1464 non-null   float64
13   Job Satisfaction        1458 non-null   object
14   Monthly Income (USD)     1470 non-null   float64
15   Salary Hike (%)          1470 non-null   float64
16   Stock Option Level       1470 non-null   float64
17   Over Time                1470 non-null   object
18   No. of Companies Worked  1470 non-null   float64
19   Total Working Years      1451 non-null   float64
20   Years At Company         1470 non-null   float64
21   Years In Current Role    1470 non-null   float64
22   Years Since Last Promotion 1470 non-null   float64
23   Years With Curr Manager  1470 non-null   float64
24   Environment Satisfaction 1470 non-null   object
25   Training Times Last Year 664 non-null    float64
26   Work life Balance        1470 non-null   object
27   Relationship Satisfaction 1470 non-null   object
28   Attrition (Yes/No)       1470 non-null   object
dtypes: float64(14), object(15)
memory usage: 333.7+ KB
```

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Files

sample_data
Employee-turnover (1) (2).xlsx

```
[7] df_null = pd.DataFrame(df.isnull().sum(), columns=["null values"])  
df_null
```

	null values
Employee Count	0
Employee ID	0
Department	0
Job Role	0
Gender	0
Age	0
Marital Status	0
Education	0
Education Field	0
Business Travel	0
Distance From Home (kms)	31
Job Involvement	0
Job Level	6
Job Satisfaction	12
Monthly Income (USD)	0
Salary Hike (%)	0
Stock Option Level	0
Over Time	0

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Files

sample_data
Employee-turnover (1) (2).xlsx

```
1-[]  
for i in df_null["null values"]:  
    1.append(1/len(df))  
df_null["Ratio of null values to total values"]= 1  
df_null
```

	null values	Ratio of null values to total values
Employee Count	0	0.000000
Employee ID	0	0.000000
Department	0	0.000000
Job Role	0	0.000000
Gender	0	0.000000
Age	0	0.000000
Marital Status	0	0.000000
Education	0	0.000000
Education Field	0	0.000000
Business Travel	0	0.000000
Distance From Home (kms)	31	0.021088
Job Involvement	0	0.000000
Job Level	6	0.004082
Job Satisfaction	12	0.008163
Monthly Income (USD)	0	0.000000
Salary Hike (%)	0	0.000000
Stock Option Level	0	0.000000

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+ Code + Text

```
df.drop(columns="Training Times Last Year")
```

	Employee Count	Employee ID	Department	Job Role	Gender	Age	Marital Status	Education	Education Field	Business Travel	No. of Companies Worked	Total Working Years	Years At Company	Years In Current Role
0	1.0	1.0	Sales	Sales Executive	Female	41.0	Single	College	Life Sciences	Travel_Rarely	2.0	8.0	6.0	4.0
1	1.0	2.0	Research & Development	Research Scientist	Male	49.0	Married	Below College	Life Sciences	Travel_Frequently	1.0	10.0	10.0	7.0
2	1.0	4.0	Research & Development	Laboratory Technician	Male	37.0	Single	College	Other	Travel_Rarely	6.0	7.0	0.0	0.0
3	1.0	5.0	Research & Development	Research Scientist	Female	33.0	Married	Master	Life Sciences	Travel_Frequently	1.0	8.0	8.0	7.0
4	1.0	7.0	Research & Development	Laboratory Technician	Male	27.0	Married	Below College	Medical	Travel_Rarely	9.0	6.0	2.0	2.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
1465	1.0	2061.0	Research & Development	Laboratory Technician	Male	36.0	Married	College	Medical	Travel_Frequently	4.0	17.0	5.0	2.0
1466	1.0	2062.0	Research & Development	Healthcare Representative	Male	39.0	Married	Below College	Medical	Travel_Rarely	4.0	9.0	7.0	7.0
1467	1.0	2064.0	Research & Development	Manufacturing Director	Male	27.0	Married	Bachelor	Life Sciences	Travel_Rarely	1.0	6.0	6.0	2.0
1468	1.0	2065.0	Sales	Sales Executive	Male	49.0	Married	Bachelor	Medical	Travel_Frequently	2.0	17.0	9.0	6.0
1469	1.0	2068.0	Research & Development	Laboratory Technician	Male	34.0	Married	Bachelor	Medical	Travel_Rarely	2.0	6.0	4.0	3.0

1470 rows x 28 columns

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Files

- sample_data
- Employee-turnover (1) (2).xlsx

+ Code + Text

```
[10] df.rename(columns={"Education Field": "Field of Education", inplace = True)
```

```
[11] print(df["Distance From Home (kms)"].describe())
```

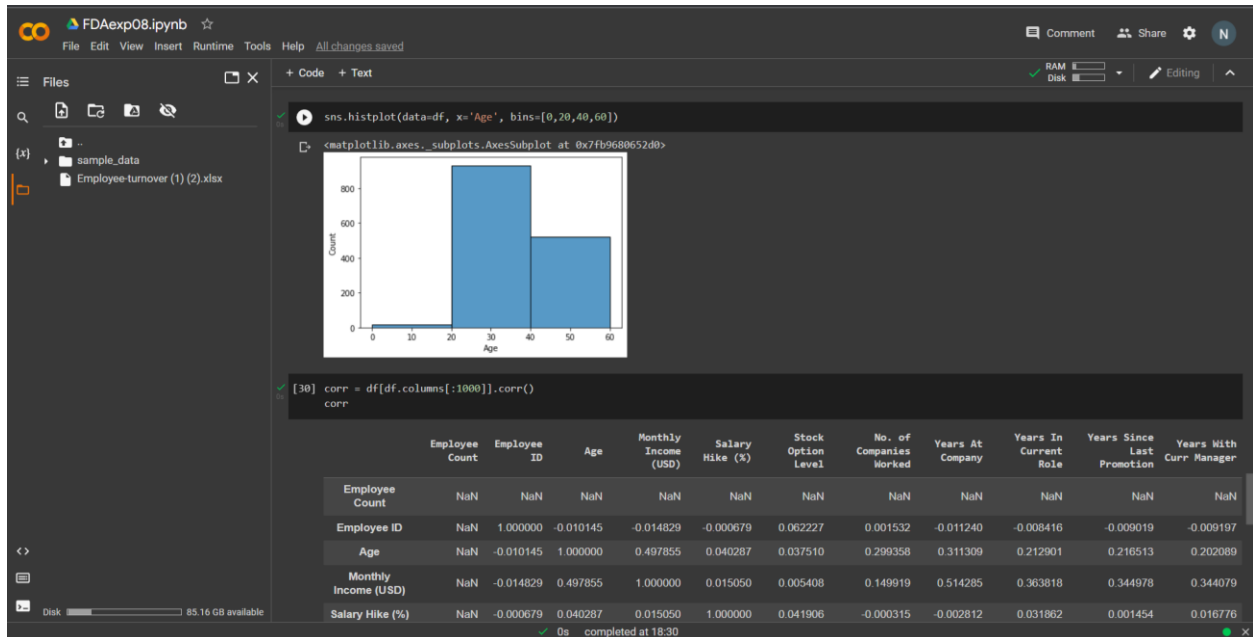
```
count    1439.0  
unique      38.0  
top         2.0  
freq      208.0  
Name: Distance From Home (kms), dtype: float64
```

```
df["Distance From Home (kms)"].fillna(0, inplace=True)
```

```
[13] df=df.dropna(axis=1)  
df.isnull().sum()  
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 1470 entries, 0 to 1469  
Data columns (total 25 columns):  
# Column Non-Null Count Dtype  
---  
0 Employee Count 1470 non-null float64  
1 Employee ID 1470 non-null float64  
2 Department 1470 non-null object  
3 Job Role 1470 non-null object  
4 Gender 1470 non-null object  
5 Age 1470 non-null float64  
6 Marital Status 1470 non-null object  
7 Education 1470 non-null object  
8 Field of Education 1470 non-null object  
9 Business Travel 1470 non-null object  
10 Distance From Home (kms) 1470 non-null object  
11 Job Involvement 1470 non-null object  
12 Monthly Income (USD) 1470 non-null float64
```

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FDAexp08.ipynb

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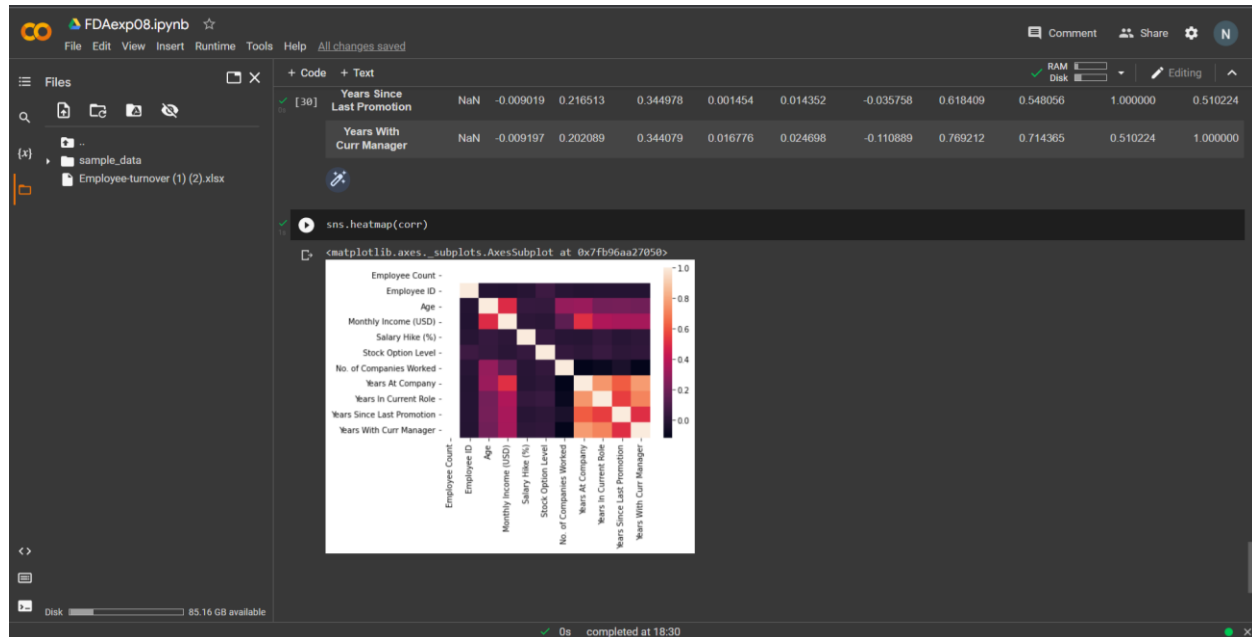
Files

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```
corr = df[df.columns[:1000]].corr()
corr
```

	Employee Count	Employee ID	Age	Monthly Income (USD)	Salary Hike (%)	Stock Option Level	No. of Companies Worked	Years At Company	Years In Current Role	Years Since Last Promotion	Years With Curr Manager
Employee Count	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Employee ID	NaN	1.000000	-0.010145	-0.014829	-0.000679	0.062227	0.001532	-0.011240	-0.008416	-0.009019	-0.009197
Age	NaN	-0.010145	1.000000	0.497855	0.040287	0.037510	0.299358	0.311309	0.212901	0.216513	0.202089
Monthly Income (USD)	NaN	-0.014829	0.497855	1.000000	0.015050	0.005408	0.149919	0.514285	0.363818	0.344978	0.344079
Salary Hike (%)	NaN	-0.000679	0.040287	0.015050	1.000000	0.041906	-0.000315	-0.002812	0.031862	0.001454	0.016776
Stock Option Level	NaN	0.062227	0.037510	0.005408	0.041906	1.000000	0.031647	0.015058	0.050818	0.014352	0.024698
No. of Companies Worked	NaN	0.001532	0.299358	0.149919	-0.000315	0.031647	1.000000	-0.118331	-0.090788	-0.035758	-0.110889
Years At Company	NaN	-0.011240	0.311309	0.514285	-0.002812	0.015058	-0.118331	1.000000	0.758754	0.618409	0.769212
Years In Current Role	NaN	-0.008416	0.212901	0.363818	0.031862	0.050818	-0.090788	0.758754	1.000000	0.548056	0.714365
Years Since Last Promotion	NaN	-0.009019	0.216513	0.344978	0.001454	0.014352	-0.035758	0.618409	0.548056	1.000000	0.510224
Years With Curr Manager	NaN	-0.009197	0.202089	0.344079	0.016776	0.024698	-0.110889	0.769212	0.714365	0.510224	1.000000

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- The Attribute Years at current Role and years with current manager
- The Attribute Years at current Role and years at a company have very high correlation.

Conclusion: Data preprocessing has been performed on the given dataset by dropping & replacing the null values. Questions related to the experiment have been answered.